

MAS Architecture and Evaluation Report

Executive Summary:

The report summarizes the multi-agent system (MAS) architecture and evaluation using the full benchmark data.

System Architecture Overview:

The following table outlines the agents used for the report and the associated functions and tools created to fulfill each of the requirements and product the final output. The functions are listed as:

Agents	Description	Structure/Functions
Orchestrator	Central router that manages the flow end-to-end, it goes as follows: request -> parsing -> inventory and ETA -> quoting. Manages other agents.	Inventory_agent Quoting_agent Sales_agent
OrderProcessorAgent	Interprets the order into a structured bill of materials (BOM) and delivery dates. Outputs a single consolidated	Extract_request_date(text) Count_order_quantities()

	BOM normalizes item names and flag items that are unavailable	
InventoryManagementAgent	Examines availability and executes based on the results. This provides two possible outcomes:	Evaluate_purchase_requirement(item_name, quantity_required, input_date_str) Stock_level(item, as_of_date)
QuotingAgent	Generates a total price for the BOM by multiplying quantities by the unit_price for each product.	Generate_quotation(bill_of_materials) Search_quote_history(terms) Quotes, quote_requests
ReportingAgent	Composes point-in- time cash balance, inventory valuation, total assets, inventory breakdown and top- selling products.	Generate_financial_report(as_of_date) Get_cash_balance()

Evaluation Results:

Total requests	20
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Requests fulfilled	10 (associated IDs: 1,4,5,10,11,13,14,15,16,18)
Unfulfilled requests	10 (associated IDs: 2,3,6,7,8,9,12,17,19,20)
Cash Balance changes found	5

Strengths:

- Functions aligned with rubric expectations.
- Additional helper functions to outline functions outside of the main agent functions to support the overall results

Areas for Improvement:

- Possible double writes
- Substitutions are hard-coded as part of the catalog, a longer inventory list will require a different operational approach